

Strategic Domain Selection and Ambiguity-Aware Mitotic Figure Detection

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Goal

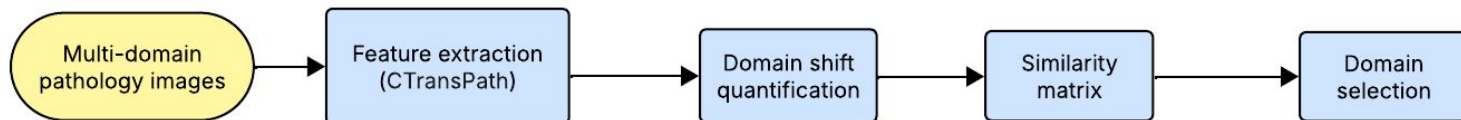
Develop an automated detection system that performs consistently well across a variety of clinical settings.

 Phase 1: Quantitative Domain Analysis & Strategic Selection

 Phase 2: U-Net segmentation & Classification Pipeline

 Phase 3: PHH3 Ambiguity Modeling & Uncertainty Quantification

Proposed Method - Phase 1: Domain Analysis Foundation



- **Feature Extraction**

- CTransPath: pre-trained feature extractor
- UMAP: Visualize feature distribution
 - Visualize domain clusters

- **Domain shift quantification**

- MMD, CORAL, Proxy A-distance, and Wasserstein

- **Similarity matrix**

- Aggregate multi-metric results

- **Domain selection**

- Select the most similar training domain

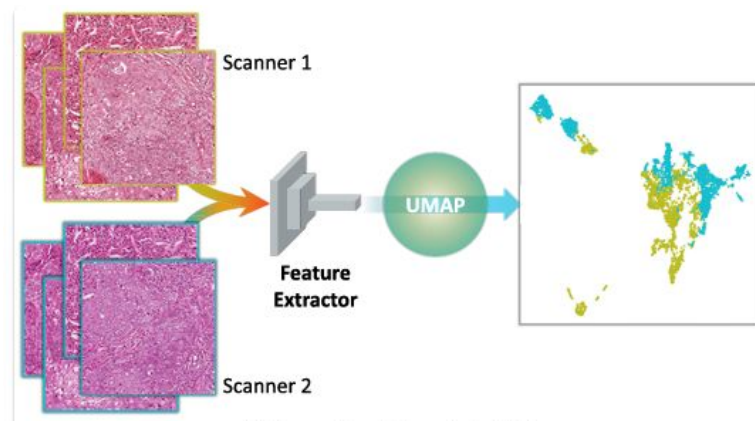


Figure 1: Feature extractor and UMAP process [4]

Addressing Bias in Domain Analysis

Potential Sources of Bias

Architecture Bias:

- CTransPath

Preprocessing Bias (non-biological):

- No Normalization

Experimental Design: Bias Evaluation with Comparisons

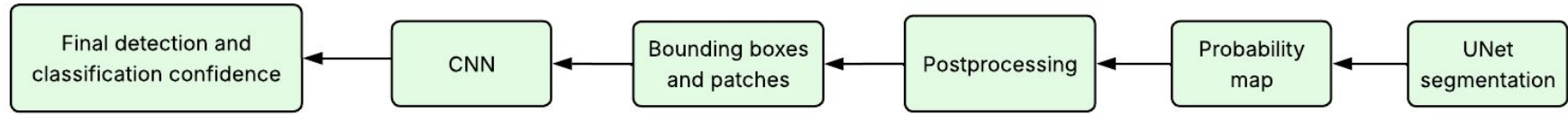
Feature Extractors:

- CTransPath (Specialized) vs. EfficientNet-B4 (General-purpose)

Preprocessing Conditions:

- Raw H&E vs. Stain-Normalized (Retinex)

Proposed Method - Phase 2: U-Net Segmentation System



- **Stain Augmentation:** Teaches model to be invariant to color
- **UNet:** Pixel-level segmentation of images
 - **Output:** Probability map
 - Per-pixel mitosis likelihood (0-1)
- **Post-processing:** Converts that heat map into precise bounding boxes
 - Thresholding, noise filtering, and boundary refinement
- **Bounding boxes & patches**
 - Define candidate regions
- **CNN**
 - Mitosis Figure, Atypical Mitosis Figure, Mitosis-Like Figure
- **Final detection & classification confidence**
 - Integrate CNN outputs and uncertainty scores

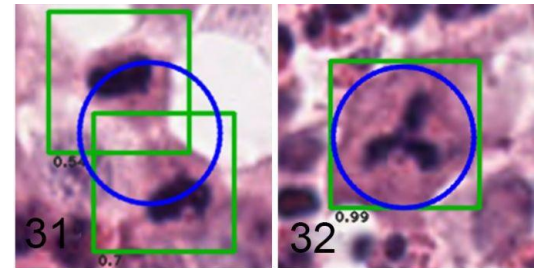
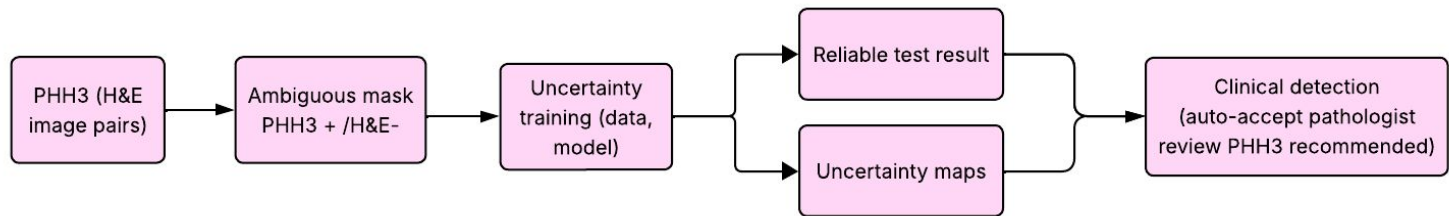


Figure 2: Boundary box (green) with confidence score and mitotic figure encircled in blue [1]

Proposed Method - Phase 3: Ambiguity Modeling with PHH3



- **PHH3**
 - PHH3-stain and H&E image pairs
 - Co-registered to identify true but ambiguous mitoses
- **Ambiguous Mask**
 - Generate binary masks for [PHH3 ] but [H&E ] = Ambiguous
- **Uncertainty training**
 - Epistemic: model uncertainty via Monte Carlo dropout
 - Aleatoric: data ambiguity from challenging morphology via heteroscedastic (variance) loss
- **Clinical detection**
 - High Confidence → Auto-accept
 - Medium Confidence → Pathologist review
 - High Uncertainty → PHH3 recommended

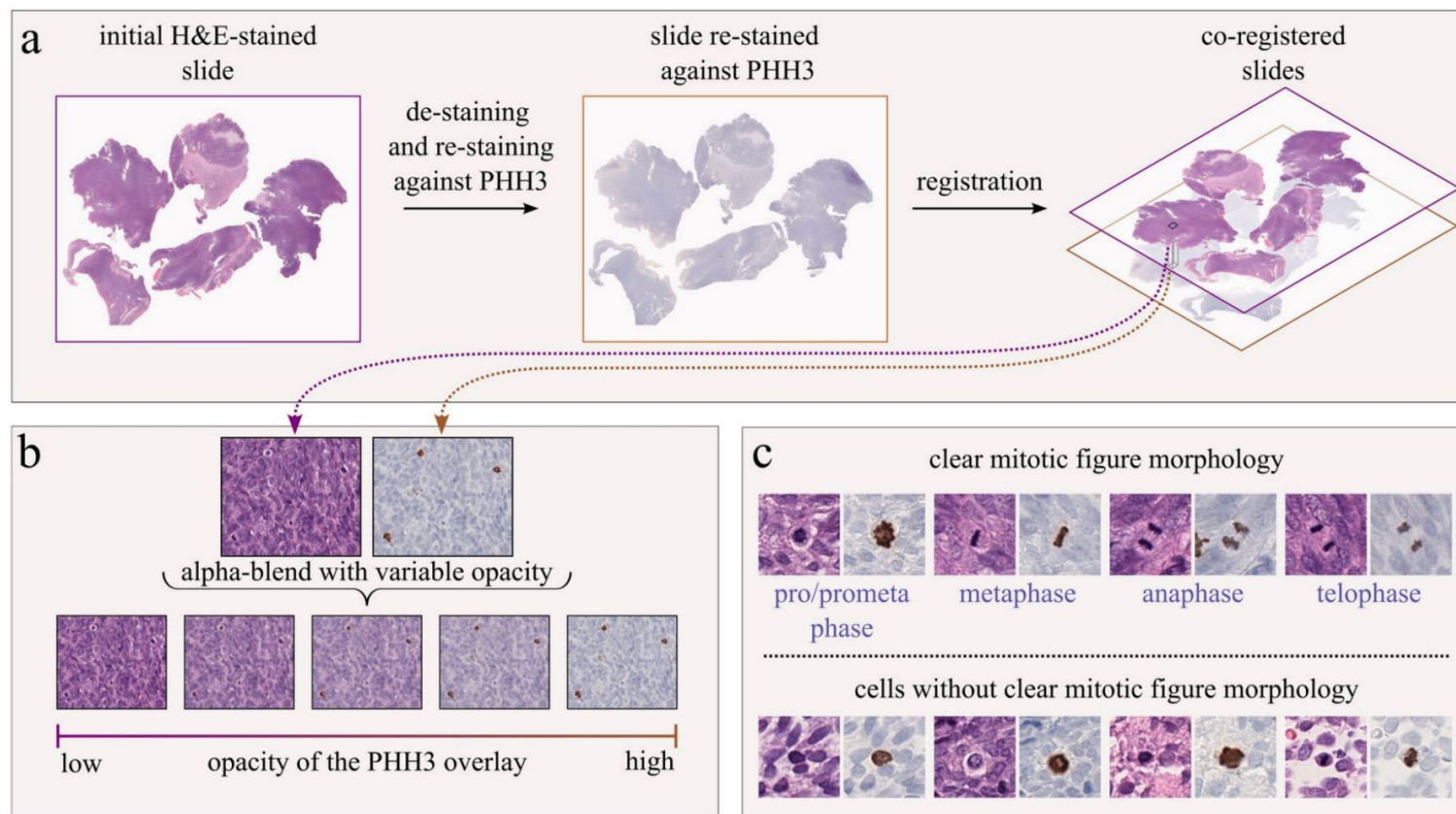


Figure 3: H&E staining procedure and results [3]

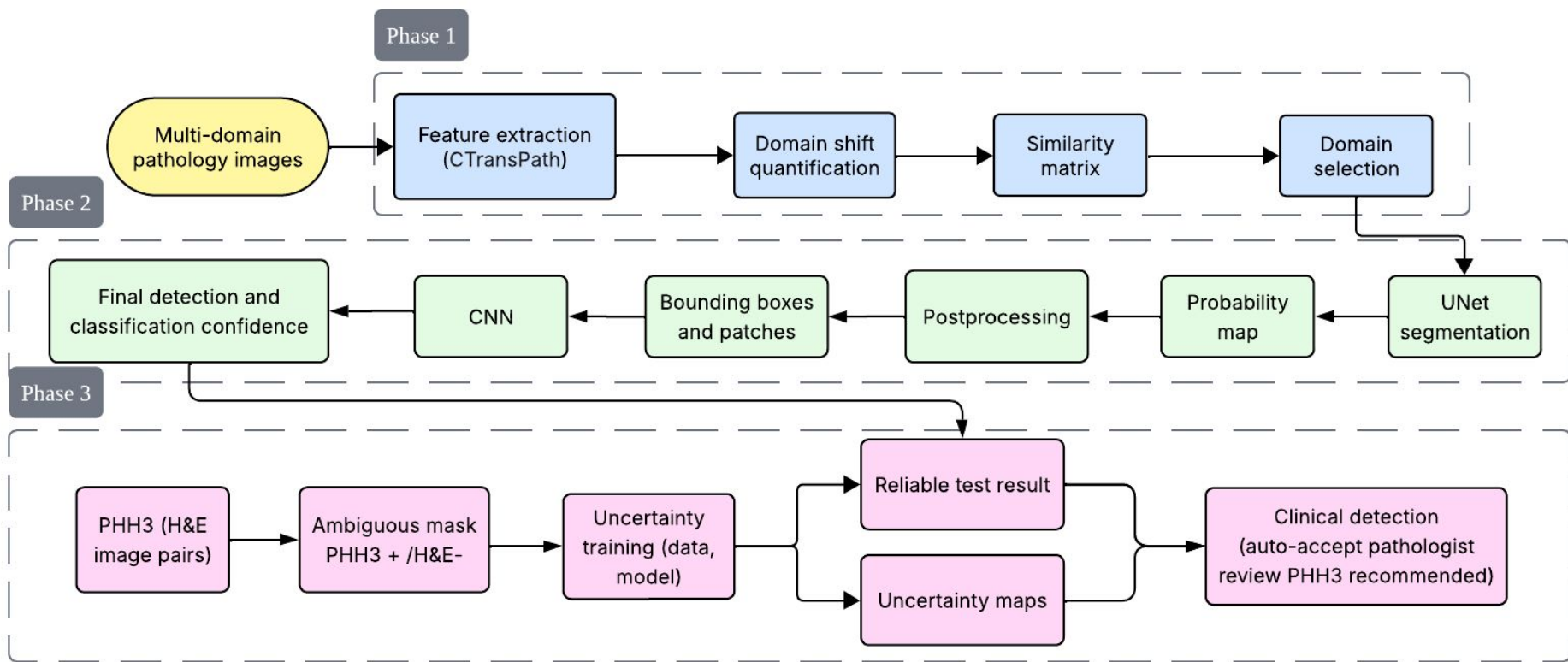


Figure 4: Planned methodological procedure including feature extraction, domain selection, processing and post-processing, and uncertainty training divided into three phases

How Our Approach Differs from Prior Work

Aspect	Prior	Our Approach
Cross-Domain Strategy	- Train on all domains - Random domain splits	- Top-k similar domain selection - Cross-species penalty quantification - Optimized domain transfer
Uncertainty Handling	- Ignore ambiguous cases - Binary predictions	- PHH3-guided ambiguity modeling - Uncertainty quantification - PHH3 masks for uncertainty-aware training

References

1. Taryn A Donovan, Frances M Moore, Christof A Bertram, Richard Luong, Pompei Bolfa, Robert Klopffleisch, Harold Tvedten, Elisa N Salas, Derick B Whitley, Marc Aubreville, et al. Mitotic figures—normal, atypical, and imposters: a guide to identification. *Veterinary pathology*, 58(2):243–257, 2021.
2. Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation, 2015.
3. Johannes Ganz et al. Information mismatch in phh3-assisted mitosis annotation leads to interpretation shifts in h&e slide analysis. *Scientific Reports*, 14:26273, 2024.
4. Mostafa Jahanifar, Manahil Raza, Kesi Xu, Trinh Thi Le Vuong, Robert Jewsbury, Adam Shephard, Neda Zamanitajeddin, Jin Tae Kwak, Shan E Ahmed Raza, Fayyaz Minhas, and Nasir Rajpoot. 2025. Domain Generalization in Computational Pathology: Survey and Guidelines. *ACM Comput. Surv.* 57, 11, Article 285 (November 2025), 37 pages.
<https://doi.org/10.1145/3724391>
5. Taryn A Donovan, Frances M Moore, Christof A Bertram, Richard Luong, Pompei Bolfa, Robert Klopffleisch, Harold Tvedten, Elisa N Salas, Derick B Whitley, Marc Aubreville, et al. Mitotic figures—normal, atypical, and imposters: a guide to identification. *Veterinary pathology*, 58(2):243–257, 2021.
6. Marc Aubreville, Nikolas Stathonikos, Taryn A Donovan, Robert Klopffleisch, Jonas Ammeling, Jonathan Ganz, Frauke Wilm, Mitko Veta, Samir Jabari, Markus Eckstein, et al. Domain generalization across tumor types, laboratories, and species—insights from the 2022 edition of the mitosis domain generalization challenge. *Medical Image Analysis*, 94:103155, 2024.
7. Xiyue Wang et al. Transformer-based unsupervised contrastive learning for histopathological image classification (ctranspath). *Medical Image Analysis*, 82:102559, 2022.

Supplementary

CTransPath

A.

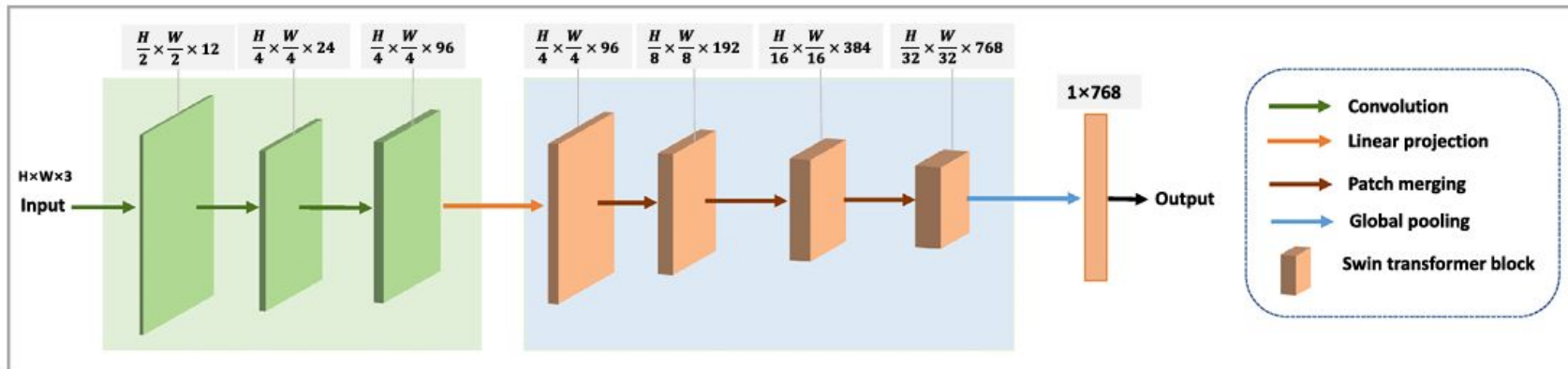


Figure 5: Hybrid CNN transformer backbone structure [7]

Standard UNet architecture

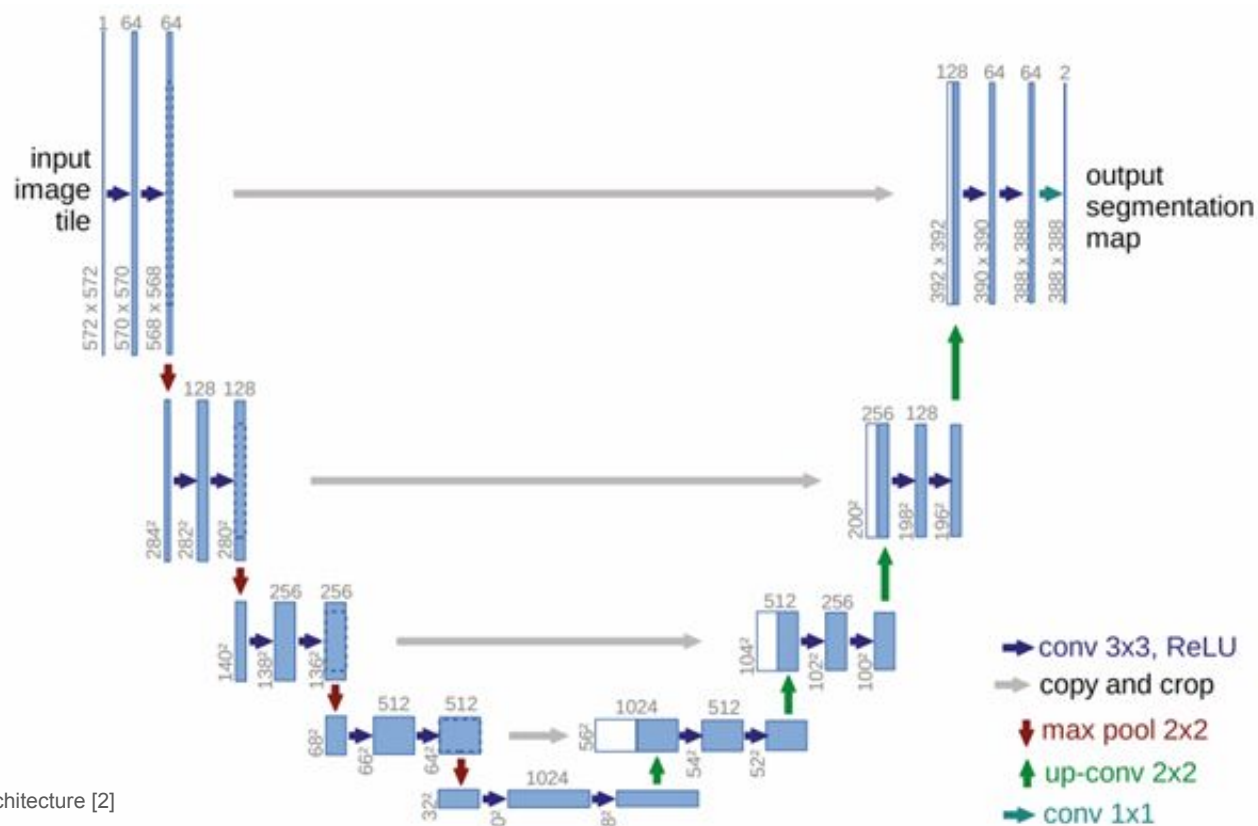


Figure 6: Standard UNet Architecture [2]

Expected outcomes

Visualization & decision support

- Confidence overlays
- Uncertainty heatmaps
- PHH3 recommendation scores
- Attention maps

Quantitative performance goals

- 50% off for F1 variance 
- +3–7 AP gain on worst domain 
- >70% auto-accept
- >90% precision

Reliability & calibration

- $ECE \leq 0.05$ 
- Confidence scores = true accuracy

Deliverable

- Open-source codebase
- Visualization dashboard

Broader Impacts

Clinical standardization

Operational efficiency

AI validation standard

Pitfalls and Mitigation

Data & domain risks

-  Inadequate feature representation →  Swap encoders + pooling + sanity check
-  Unstable metrics →  Metric ensemble + bootstrap stability

Modeling & Architecture Risks

-  Segmentation-detection interference →  Progressive training + loss balance
-  Uncertainty failure →  Combining two types of uncertainty + registration weights

Evaluation & Generalization Risks

-  Data leakage →  Strict LODO protocol
-  Cross-species gap →  Species-specific adapter or MoE

Operation risks

-  High computational overhead →  Feature caching, efficient subsampling